Professional Installation Manual

Ubuntu Server Deployment for ABC Startup Solutions

ITEP 414 – System Administration and Maintenance
Week 3 Portfolio Project

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1. Introduction

This manual documents the deployment of Ubuntu Server LTS as the first Linux server for ABC Startup Solutions. It is written so that another system administrator can reproduce the installation and configuration without needing additional instructions. The server will support file sharing, remote administration, database hosting, web hosting, and internal services.

Scope: this manual covers VM provisioning, OS installation, initial configuration, post-install verification, common troubleshooting scenarios, and operational best practices.

2. Ubuntu Server Installation Guide

2.1 Prerequisites

- Oracle VirtualBox or VMware Workstation installed on the host machine
- Ubuntu Server LTS ISO (latest stable release) downloaded
- Minimum host resources to support a 4 GB / 2 vCPU / 40 GB guest VM

2.2 Virtual Machine Specifications

Component	**Value**
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	80 GB (VDI/VMDK)
Network	NAT

2.3 Installation Steps

- Boot the VM from the Ubuntu Server ISO and select Install Ubuntu Server.
- Language: English.
- Keyboard layout: English (US).
- Network: accept DHCP; record the assigned IP address once online.
- Hostname: server01.
- Profile setup: create a non-root administrative user with a strong password.
- Disk partitioning: choose Guided – Use Entire Disk.
- Enable the OpenSSH Server feature so the server is remotely manageable on first boot.
- Skip additional snap/server packages unless specifically required.
- Complete installation, reboot, and remove the ISO from the virtual optical drive.

3. Initial Server Configuration

Setting	Value
Hostname	server01
Primary admin user	[yourlastname]
Network mode	NAT (DHCP)
SSH access	Enabled (OpenSSH Server)

After first login, apply outstanding OS patches immediately so the server starts from a fully updated baseline:

```
sudo apt update sudo apt upgrade -y
```

Confirm SSH is enabled for remote administration so future maintenance does not require console access:

```
systemctl status ssh
```

4. Server Verification Procedures

#	Check	Command	Expected Result
1	Login	(console login)	Successful login with created account
2	Hostname	hostname	server01
3	IP address	ip addr	Assigned DHCP address visible on interface
4	Internet connectivity	ping -c 4 google.com	4 packets received, 0% loss
5	System update	sudo apt update && sudo apt upgrade -y	Packages fetched/upgraded without errors
6	SSH service	systemctl status ssh	active (running)

5. Troubleshooting Guide

Symptom	Likely Cause	Resolution
VM does not obtain an IP address	Network adapter not attached, or DHCP unavailable on the selected network mode	Set the adapter to NAT (or Bridged if instructed), restart the VM, re-run 'ip addr' to confirm
SSH connection refused from host	OpenSSH Server not installed, or service not started	Run 'sudo apt install openssh-server' if missing, then 'sudo systemctl enable --now ssh'
ping to google.com fails but IP is assigned	DNS resolution issue or NAT misconfiguration	Test with 'ping -c 4 8.8.8.8'; if that works, check /etc/resolv.conf for a valid nameserver
'sudo apt upgrade' hangs or fails	Stale package lists or interrupted previous run	Run 'sudo apt update' again, then 'sudo dpkg --configure -a' before retrying the upgrade
Cannot log in with created account	Caps Lock or keyboard layout mismatch during password entry	Verify keyboard layout matches what was set during install; try a console login instead of SSH first

6. Best Practices

- Always create and use a named administrative account instead of enabling direct root login.
- Apply OS updates immediately after installation, then schedule regular update windows.
- Keep the server minimal — only install packages that are actually required for its role.
- Document the assigned hostname and IP address as soon as they are known, for future reference.
- Enable SSH key-based authentication and disable password authentication once key access is confirmed working, for production servers.
- Take a snapshot of the VM immediately after a clean, verified installation, so it can be quickly restored if configuration changes go wrong.

- Record every configuration change and the reasoning behind it, so the deployment can be reproduced or audited later.

7. References

- Ubuntu Server Documentation – https://ubuntu.com/server/docs
- OpenSSH Documentation – https://www.openssh.com/manual.html
- UEFI Specification – https://uefi.org/specifications